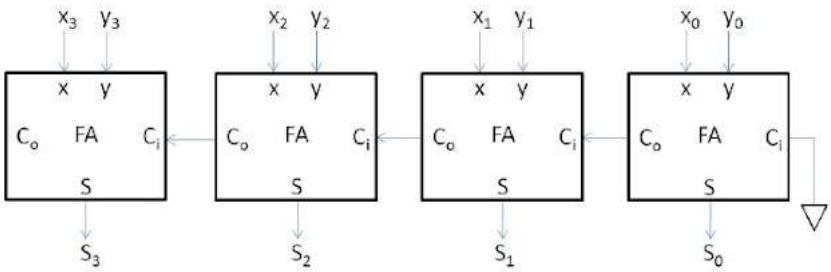
# Verilog Code for Ripple Carry Adder

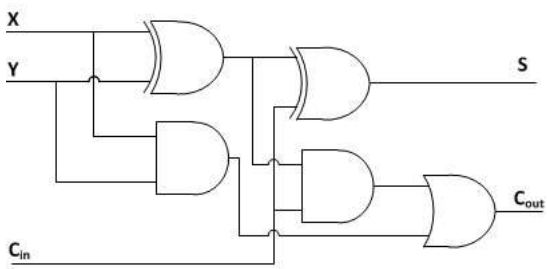
A Verilog code for a 4-bit Ripple-Carry Adder is provided in this project.

The 4-bit ripple-carry adder is built using 4 1-bit full adders as shown in the following figure.




You can find the behavioral Verilog code for 1-bit full adder: here

Or use the structural Verilog code for the full adder based on its logic diagram as follows:




Verilog code for 1-bit full adder using structural modeling:

```
// fpga4student.com: FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for 4-bit ripple-carry adder
// Verilog code for 1-bit full adder
module fulladder(X, Y, Ci, S, Co);
  input X, Y, Ci;
  output S, Co;
  wire w1,w2,w3;
  //Structural code for one bit full adder
  xor G1(w1, X, Y);
  xor G2(S, w1, Ci);
  and G3(w2, w1, Ci);
  and G4(w3, X, Y);
  or G5(Co, w2, w3);
endmodule
```

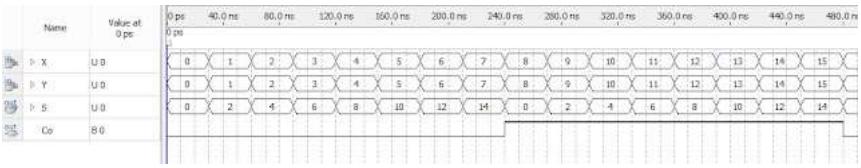
Then, instantiate the full adders in a Verilog module to create a 4-bit ripple-carry adder using structural modeling.

Following is the Verilog code for the 4-bit ripple-carry adder:

```
// fpga4student.com: FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for 4-bit ripple-carry adder
module rippe_adder(X, Y, S, Co);
 input [3:0] X, Y;// Two 4-bit inputs
 output [3:0] S;
 output Co;
 wire w1, w2, w3;
 // instantiating 4 1-bit full adders in Verilog
 fulladder u1(X[0], Y[0], 1'b0, S[0], w1);
 fulladder u2(X[1], Y[1], w1, S[1], w2);
 fulladder u3(X[2], Y[2], w2, S[2], w3);
 fulladder u4(X[3], Y[3], w3, S[3], Co);
endmodule
```

Now, it's time to run a simulation to see how it works. In this Verilog project, let's use the Quartus II Waveform Editor to create test vectors and run functional simulations without a Verilog testbench. If you

want to learn how to run the simulation without a Verilog testbench, you can check the tutorial: here.
Below is the simulation waveform for the Ripple-Carry Adder in Verilog:




The simulation waveform demonstrates the accurate functional operations of the 4-bit ripple-carry adder that is implemented in Verilog above.
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